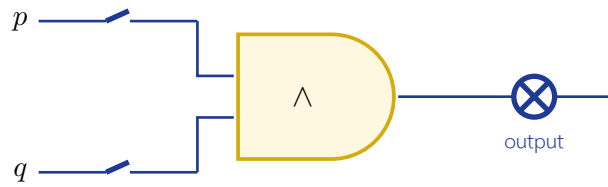
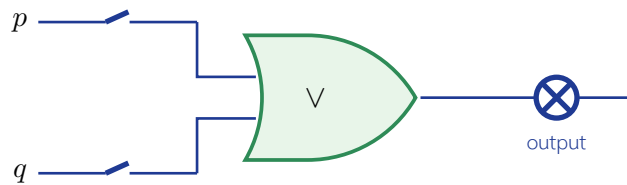


02-and-gate



p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

03-or-gate



p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

04-implication

$$p \rightarrow q$$

(อ่านว่า “ถ้า p แล้ว q ”)

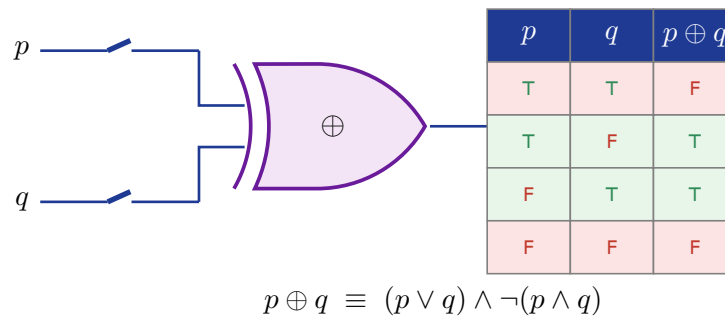
p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

เป็นเท็จ กรณีเดียว
เมื่อ p จริง แต่ q เท็จ

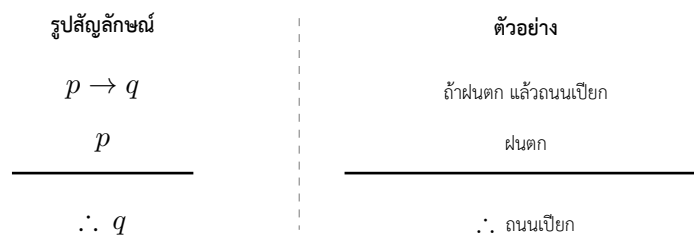
คิดเป็นคำสัญญา: “ถ้าฝนตก แล้วถนนเปียก”

คำสัญญานี้จะ **ผิด (เท็จ)** ก็ต่อเมื่อ “ฝนตก แต่ถนนไม่เปียก” เท่านั้น ส่วนอีก 3 กรณีถือว่า **รักษาสัญญา (จริง)**

09-xor

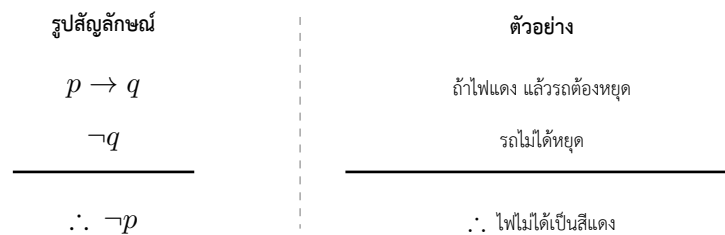


07-modus-ponens



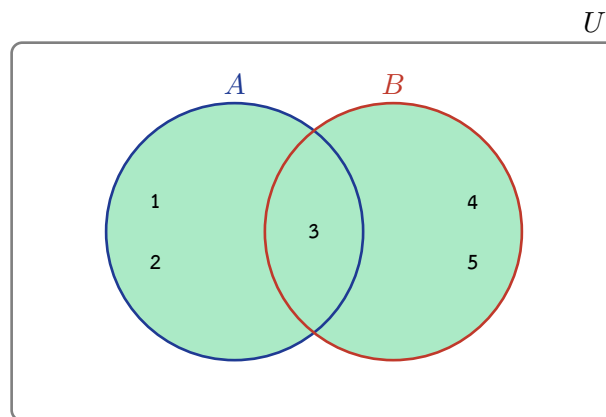
กฎ Modus Ponens: เมื่อ " $p \rightarrow q$ " และ " p " เป็นจริงทั้งคู่ จึงสรุปได้ว่า " q " เป็นจริง

08-modus-tollens



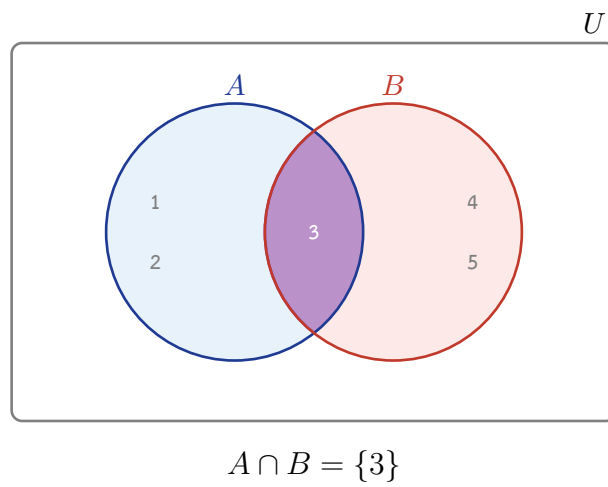
กฎ Modus Tollens: เมื่อ " $p \rightarrow q$ " เป็นจริง และ " q " เป็นเท็จ ($\neg q$) จึงสรุปได้ว่า " p " เป็นเท็จ ($\neg p$)

10-union

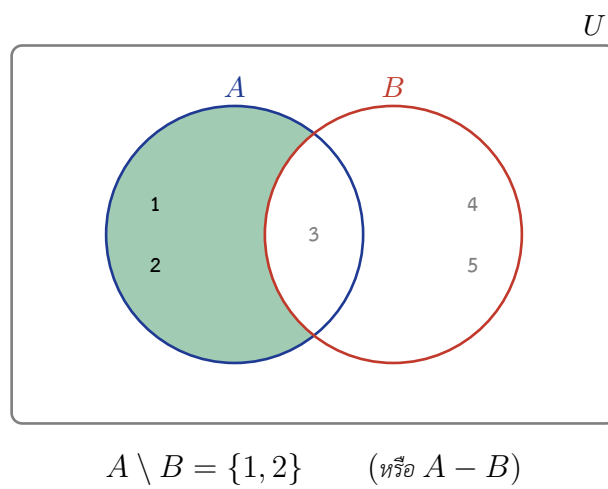


$$A \cup B = \{1, 2, 3, 4, 5\}$$

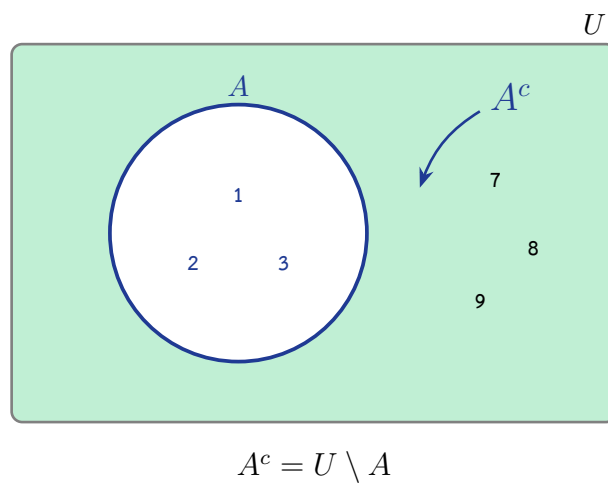
11-intersection



13-difference

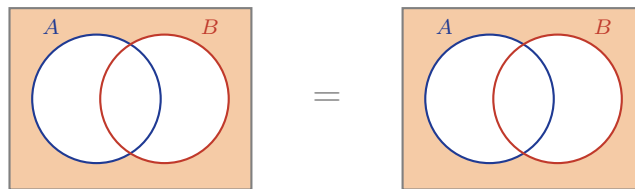


14-complement

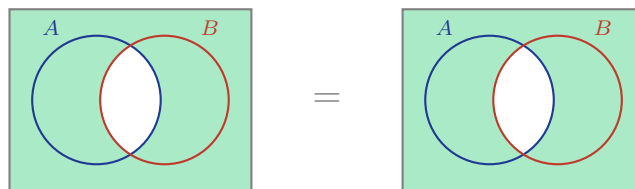


15-demorgan-sets

$$(A \cup B)^c = A^c \cap B^c$$

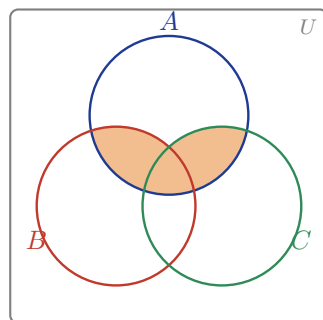


$$(A \cap B)^c = A^c \cup B^c$$



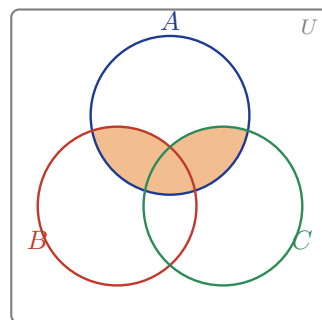
16-distributive

$$A \cap (B \cup C)$$

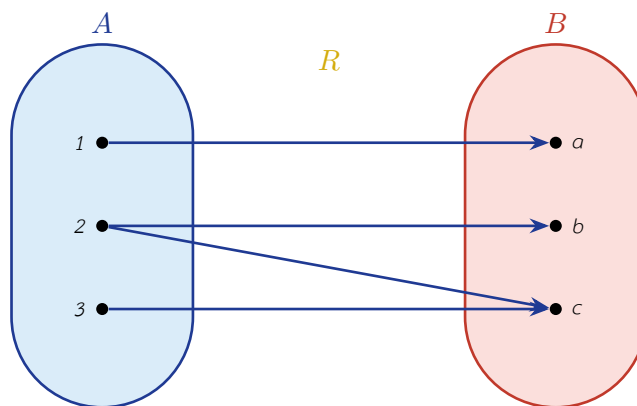


✓
=

$$(A \cap B) \cup (A \cap C)$$

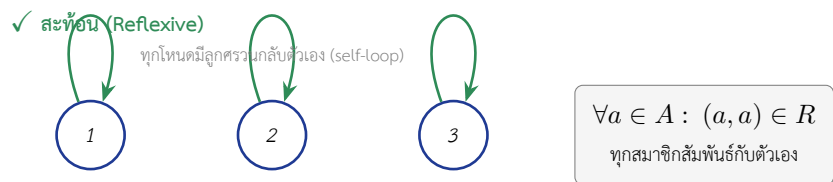


18-relation-graph

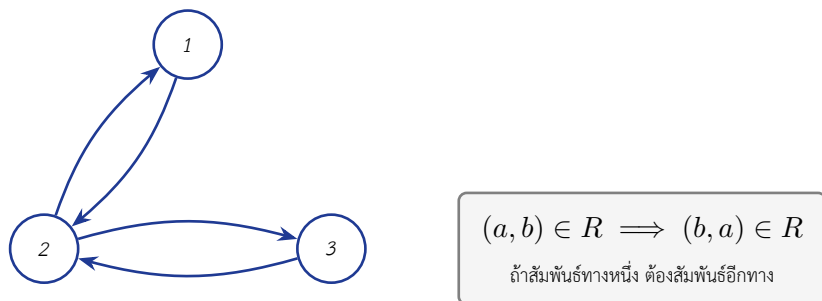


$$R = \{(1, a), (2, b), (2, c), (3, c)\} \subseteq A \times B$$

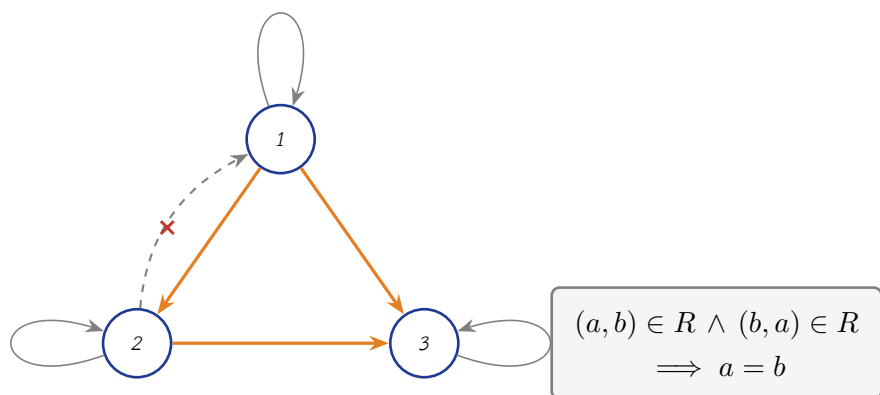
19-reflexive



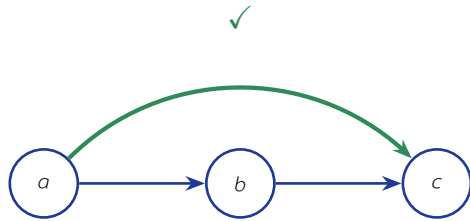
20-symmetric



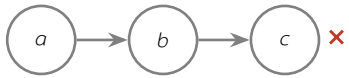
21-antisymmetric



22-transitive



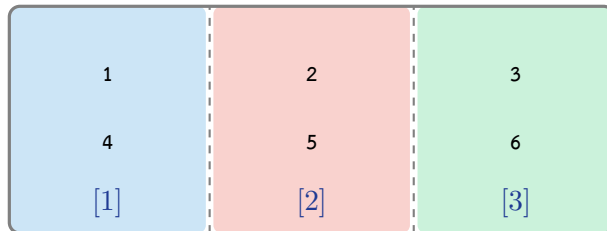
$$a R b \wedge b R c \implies a R c$$



ไม่ถ่ายทอด

23-equivalence

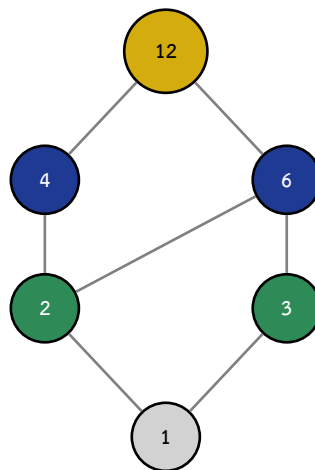
$$U = \{1, 2, 3, 4, 5, 6\}$$



- ✓ สะท้อน (Reflexive)
- ✓ สมมาตร (Symmetric)
- ✓ ถ่ายทอด (Transitive)

$$A/R = \{[1], [2], [3]\}$$

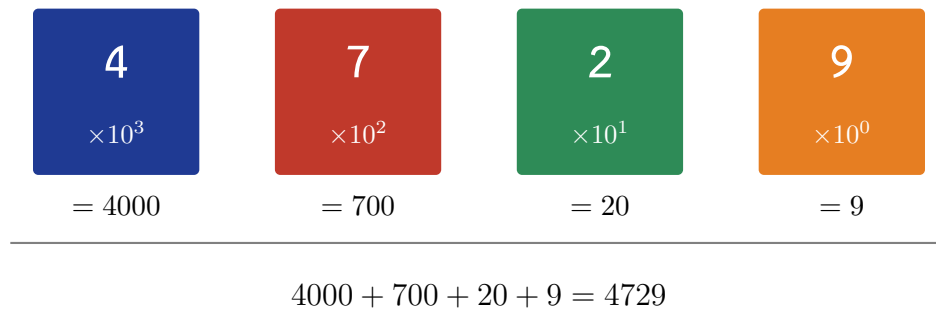
24-hasse



$$a \mid b$$

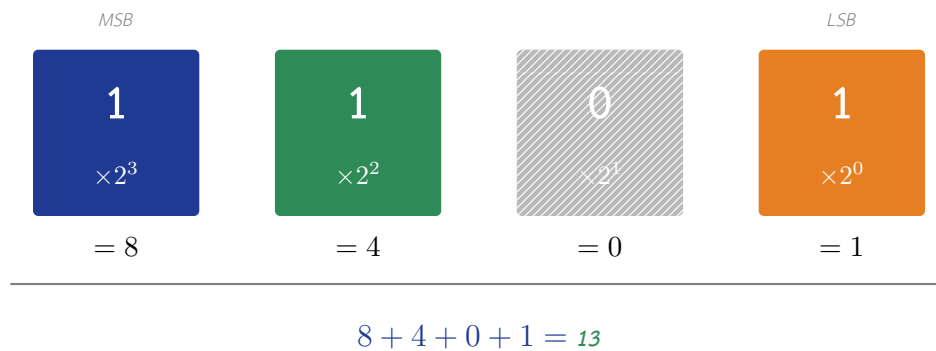
25-decimal-pv

4729

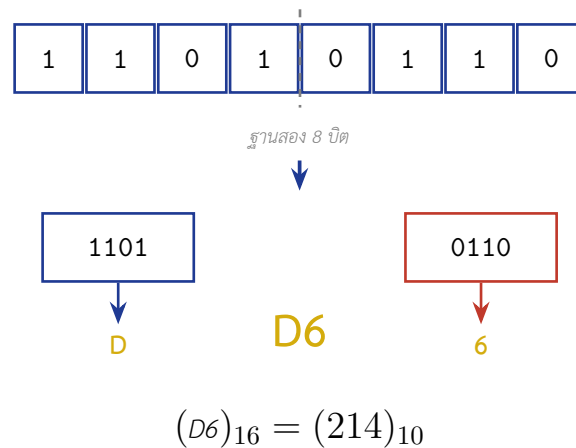


26-binary-pv

$$(1101)_2 = (13)_{10}$$

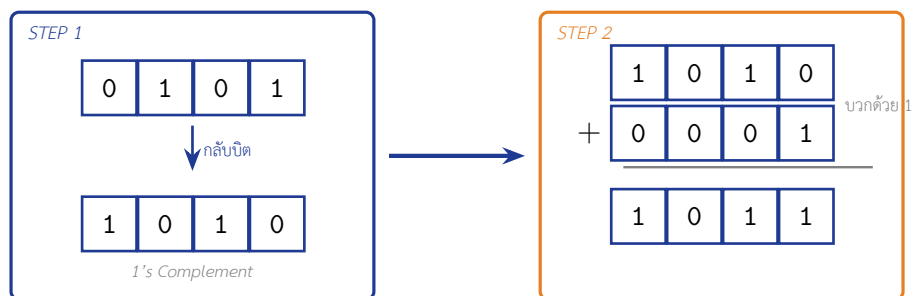


27-hex



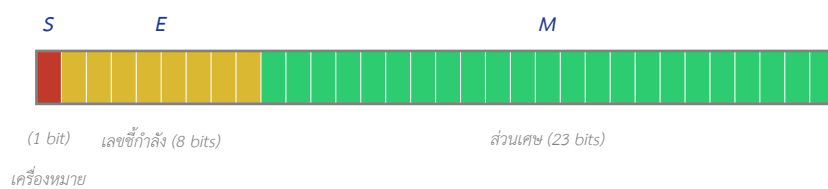
28-2s-comp

$$(0101)_2 = +5$$



$$(1011)_2 = -5$$

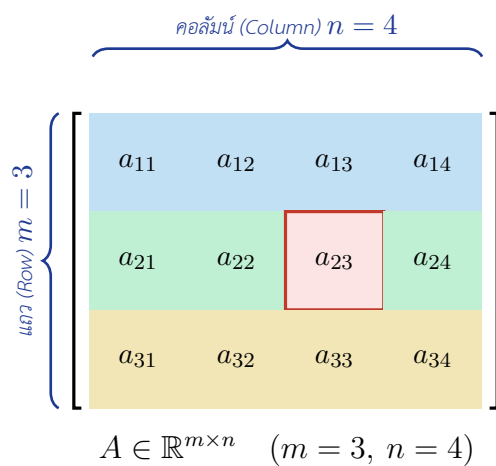
29-ieee754



$$-3.5 = 1 \mid 10000000 \mid 11000000000000000000000$$

$$value = (-1)^s \times 1.M \times 2^{(E-127)}$$

30-dimensions



31-types

<i>Zero</i> $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ เมทริกซ์ศูนย์	<i>Identity</i> $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ เมทริกซ์เอกลักษณ์	<i>Diagonal</i> $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \end{pmatrix}$ เมทริกซ์ทแยงมุม
<i>Upper Triangular</i> $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ เมทริกซ์สามเหลี่ยมบน	<i>Lower Triangular</i> $\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$ เมทริกซ์สามเหลี่ยมล่าง	<i>Symmetric</i> $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$ เมทริกซ์สมมาตร

32-multiplication

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \times \begin{pmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{pmatrix} = \begin{pmatrix} 58 & 64 \\ 139 & 154 \end{pmatrix}$$

$$c_{11} = (1)(7) + (2)(9) + (3)(11) = 58$$

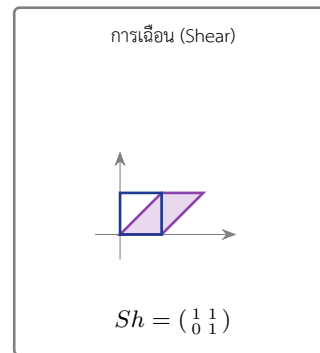
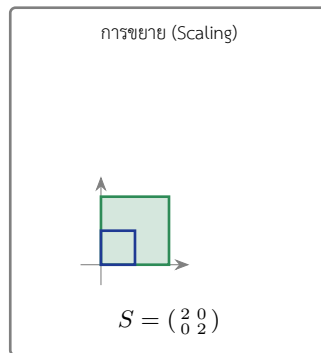
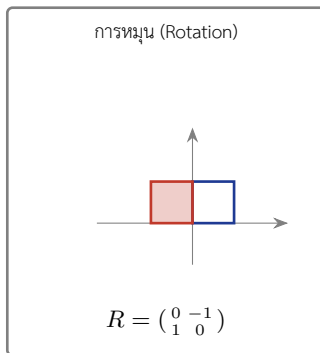
$$A_{(m \times k)} \cdot B_{(k \times n)} = C_{(m \times n)}$$

33-determinant

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \det(A) = ad - bc$$

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix} \Rightarrow \det(A) = (3)(4) - (2)(1) = 12 - 2 = 10$$

34-transformation



35-function-map

